CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2020 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define DML (Data Manipulation Language).

Definition: Data Manipulation Language (DML) provides constructs for accessing, inserting, updating, and querying stored database records. Classified into *Procedural DML* (user specifies what data and how to retrieve it) and *Non-Procedural / Declarative DML* (SQL, where user specifies only what data is needed).

Q2 (3M) Give examples of Simple and Composite Attributes of an entity.

Examples:

- **Simple (Atomic) Attribute:** Cannot be divided into subparts (e.g., `RollNo = 101`, `Age = 21`, `Salary = 50000`).
- **Composite Attribute:** Can be divided into smaller sub-components with independent meanings (e.g., `Address` divided into `HouseNo`, `Street`, `City`, `Pincode`; or `Name` into `FirstName`, `LastName`).

Q3 (3M) What do you mean by Referential Integrity Constraint?

Definition: Referential Integrity requires that any value of a foreign key attribute in a referencing relation must either match an existing primary key value in the referenced relation or be `NULL`. Enforces valid relationships and prevents "orphan records".

Q4 (3M) What is a Data Dictionary (System Catalog)?

Definition: A **Data Dictionary** is a reserved set of read-only system tables maintained by the DBMS that stores metadata (data about data), including schemas, table descriptions, column domains, integrity constraints, indexes, user privileges, and storage allocations.

Q5 (3M) Define Relational Schema.

Definition: A **Relational Schema** denotes the design structure of a relation, symbolized as R(A_1, A_2, \dots, A_n), where R is the relation name and A_i are attribute names defined over specific domains dom(A_i).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Lossless-Join Decomposition and Dependency Preservation in relational design.

1. Lossless-Join Decomposition:
A decomposition of relation R into R_1 and R_2 is lossless if and only if natural join of the projections equals the original relation without generating spurious tuples:

```
R_1 ⋈ R_2 = R ⇔ (R_1 ∩ R_2 → R_1) OR (R_1 ∩ R_2 → R_2)
```

2. Dependency Preservation:
A decomposition preserves dependencies if the union of functional dependencies enforceable within individual decomposed relations covers all functional dependencies of the original relation: (F_1 \cup F_2)^+ = F^+.

Q7 (7.5M) **Compare 3NF and BCNF with an illustrative example.**

Key Differences:

- **3NF:** For every non-trivial FD X \to Y, either X is a super key *OR* Y is a prime attribute (part of any candidate key). Allows dependency preservation always.
- **BCNF:** For every non-trivial FD X \to Y, X **must strictly be a super key**. Eliminates all FD redundancy, but may not always preserve all functional dependencies.

Q8 (7.5M) **Draw an ER Diagram for a University Examination System with entities, attributes, and relationships.**

Entities: `STUDENT` (RollNo, Name), `COURSE` (CourseCode, Title), `EXAM_CENTER` (CenterID, Location), `ADMIT_CARD` (AdmitCardNo).

```
[STUDENT] ----- (1:1) ----- [ADMIT_CARD]
    | (M)
    |
 < ENROLLS >
    |
    v (N)
 [COURSE] ----- (N:1) ----- [EXAM_CENTER]
```

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Discuss Relational Algebra fundamental operations with examples.**

Operations:

1. **Selection (σ):** \sigma_{Marks \ge 60}(Student)
2. **Projection (π):** \pi_{RollNo, Name}(Student)
3. **Cartesian Product (×):** Student \times Course
4. **Union (∪):** SectionA \cup SectionB
5. **Set Difference (−):** AllStudents - PassedStudents
6. **Rename (ρ):** \rho_{S}(Student)

Q10 (15M) **Explain File Organization Techniques: Sequential, Indexed Sequential (ISAM), and Hashed File Organization.**

1. Sequential File Organization: Records stored consecutively ordered by a search key. Fast for full table scans, slow O(N) for insertions/random lookups.
2. Indexed Sequential Access Method (ISAM): Combines sequential ordering with hierarchical index blocks (Cylinder, Track index). Enables rapid binary search.
3. Hashed File Organization: Uses hash function on search key for direct bucket addressing. Offers O(1) lookups, but poor for range queries.

Q11 (15M) **Explain Database Security and Authorization. Discuss GRANT and REVOKE commands in SQL.**

SQL Security Commands:

```
-- Granting SELECT and INSERT privileges with delegation rights
GRANT SELECT, INSERT ON Employee TO user_ravi WITH GRANT OPTION;

-- Granting role-based access
CREATE ROLE clerk;
GRANT SELECT, UPDATE(Salary) ON Employee TO clerk;
GRANT clerk TO user_amit;

-- Revoking privileges
REVOKE INSERT ON Employee FROM user_ravi CASCADE;
```

Q12 (15M) **Explain Two-Phase Locking (2PL) and describe Deadlock Prevention and Detection mechanisms.**

Deadlock Handling:

- **Deadlock Prevention:** Uses transaction timestamps to avoid circular wait:
  - *Wait-Die (Non-preemptive):* Older waits for younger; younger dies immediately.
  - *Wound-Wait (Preemptive):* Older wounds (preempts) younger; younger waits for older.
- **Deadlock Detection:** Builds periodic *Wait-For Graph (WFG)*; cycles in the graph indicate deadlocks, resolved by aborting the lowest-cost victim transaction.

Q13 (15M) **Explain Views in SQL. What are the advantages of Views? When is a View updatable?**

Views in SQL: A **View** is a virtual table defined by an underlying stored SQL query.
Advantages: Data security (restricts column access), simplified queries, and logical data independence.
Updatable View Criteria:

- Query contains only one base table in `FROM` clause.
- No aggregate functions (`SUM` , `AVG` , `COUNT`).
- No `GROUP BY` , `HAVING` , `DISTINCT` , or `UNION` .
- Primary key of base table must be included in the view columns.
